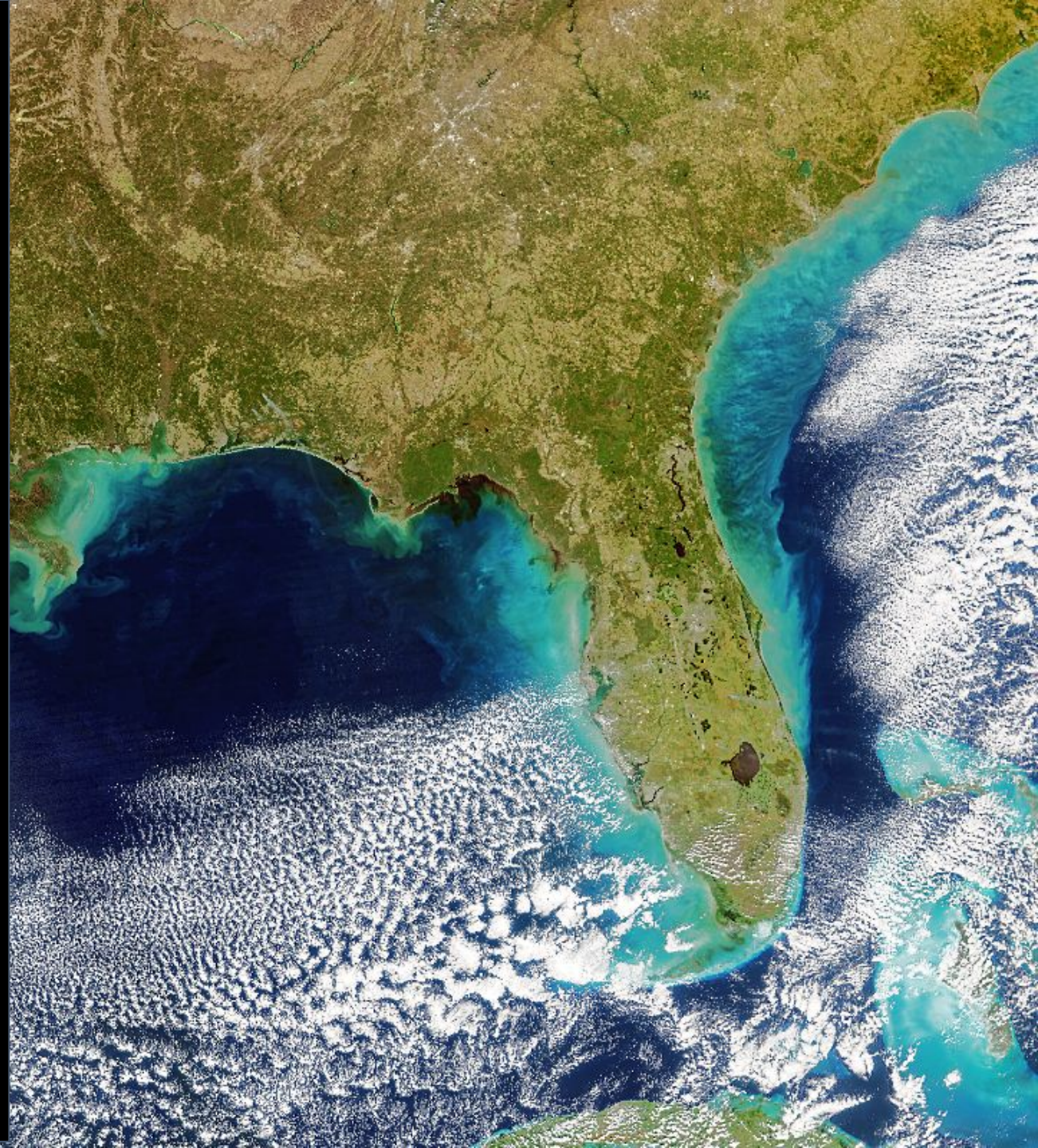


Introduction to PACE hyperspectral ocean color data for fisheries applications

Thanks to
Jeremy Werdell, PACE Project Scientist, NASA
Goddard for PACE slides
and to NASA for images



Late 2014 PACE became a mission and work began early 2015



LAUNCH 8 February 2024

Pictures from NASA

NASA Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission

The **PACE (Plankton, Aerosol, Cloud, ocean Ecosystem)** satellite mission is a NASA Earth science project aimed at studying the Earth's ocean ecosystems and atmosphere. It will collect data on the health of the oceans, the composition of aerosols in the atmosphere, and the characteristics of clouds.



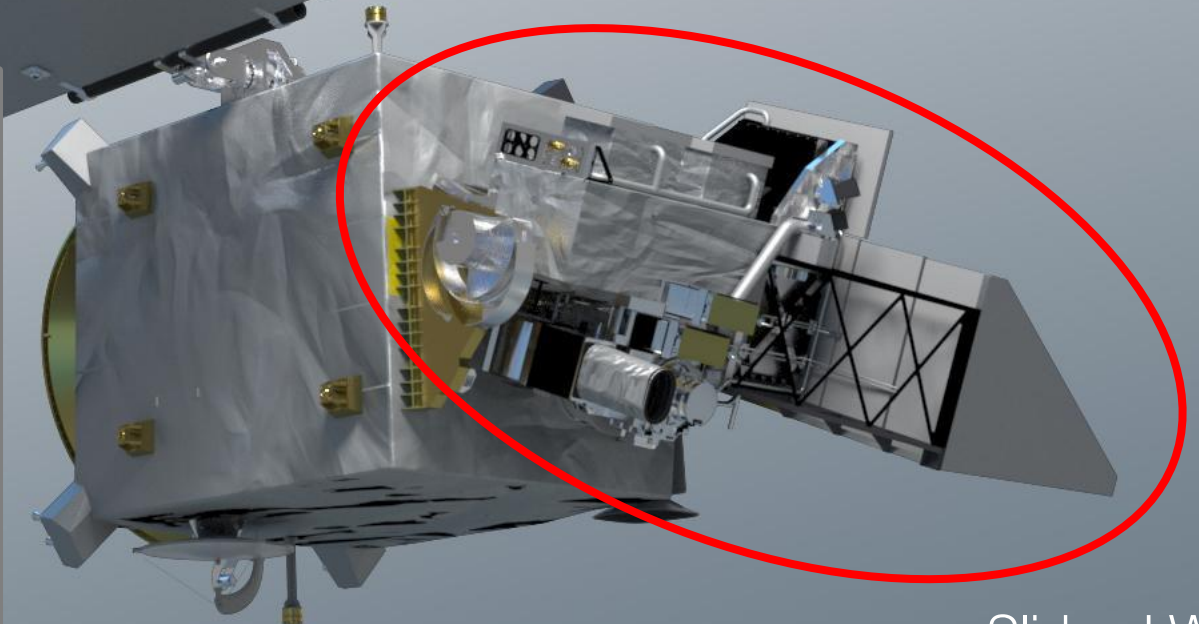
The sensors:

- Ocean Color Instrument (primary)
- SPeXone
- HARP2

ocean color & the ocean color instrument

ocean color retrievals drive OCI's
design & performance requirements

- hyperspectral scanning radiometer
- (320) 340 – 890 nm, 5 nm resolution, 2.5 nm steps⁺
- plus, 940, 1038, 1250, 1378, 1615, 2130, and 2250 nm
- *single science pixel to mitigate image striping*
- 1 – 2 day global coverage
- ground pixel size of 1 km² at nadir
- ± 20° fore/aft tilt to avoid Sun glint
- twice monthly lunar calibration
- daily on-board solar calibration
- <0.5% total system error for VIS-NIR
- SNRs optimized for ocean color science
- [simulated top-of-atmosphere data available](#)

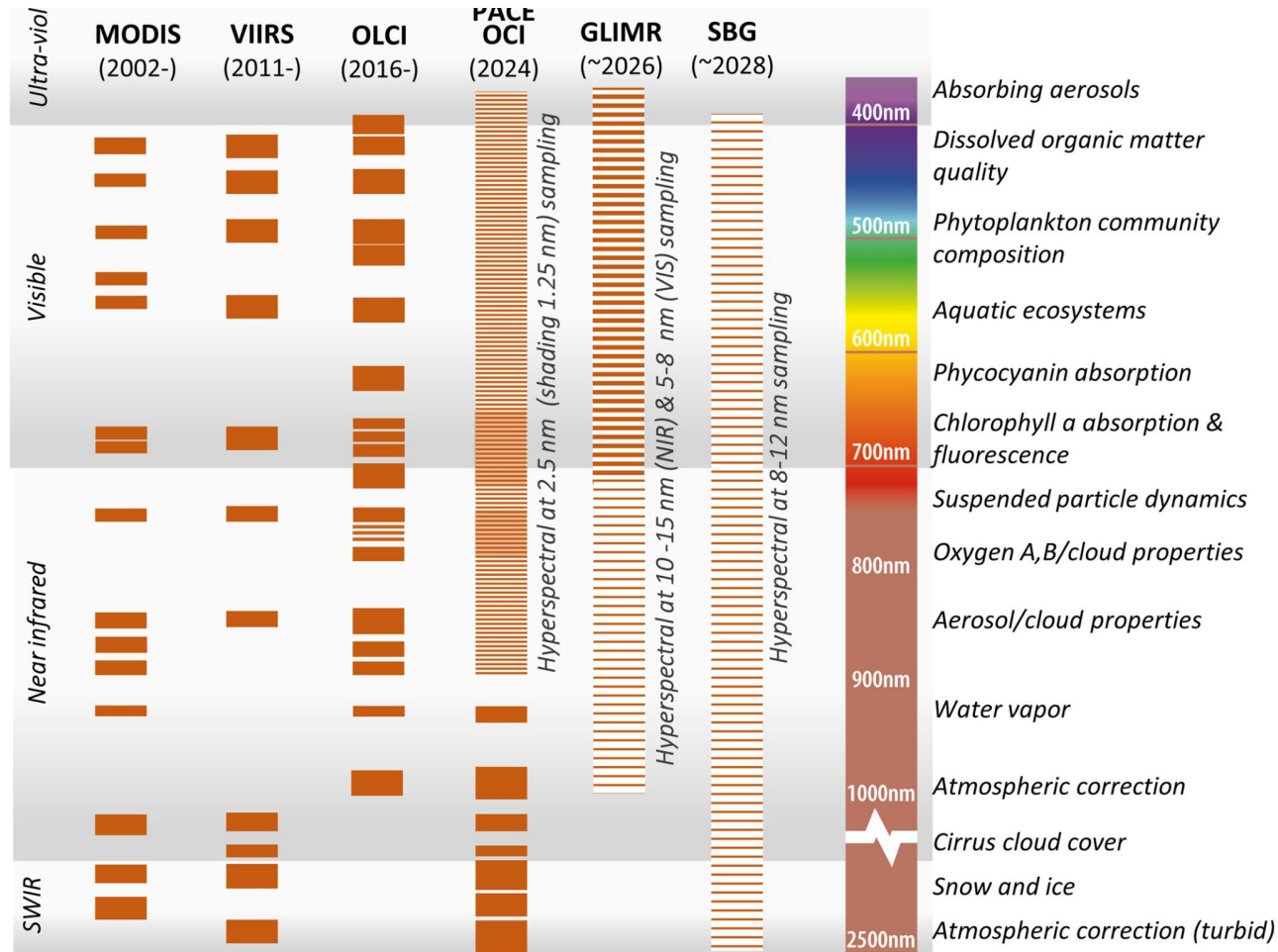


+ with 1.25 nm steps in several spectral regions

* developed primarily for mechanical processing assessments

Slide, J Werdell

Hyperspectral allows us to distinguish what is in the water

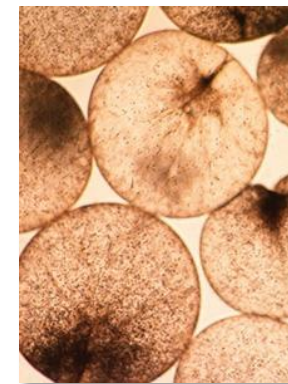


Example diatom



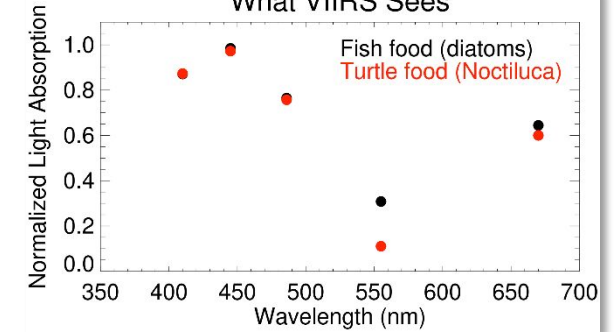
Linda Armbricht, abc.com.au

Example Noctiluca

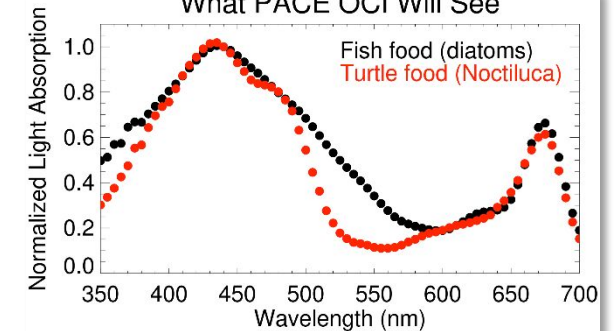


— 1 mm —
Joaquim Goes, LDEO

What VIIRS Sees

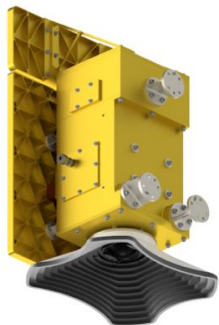


What PACE OCI Will See

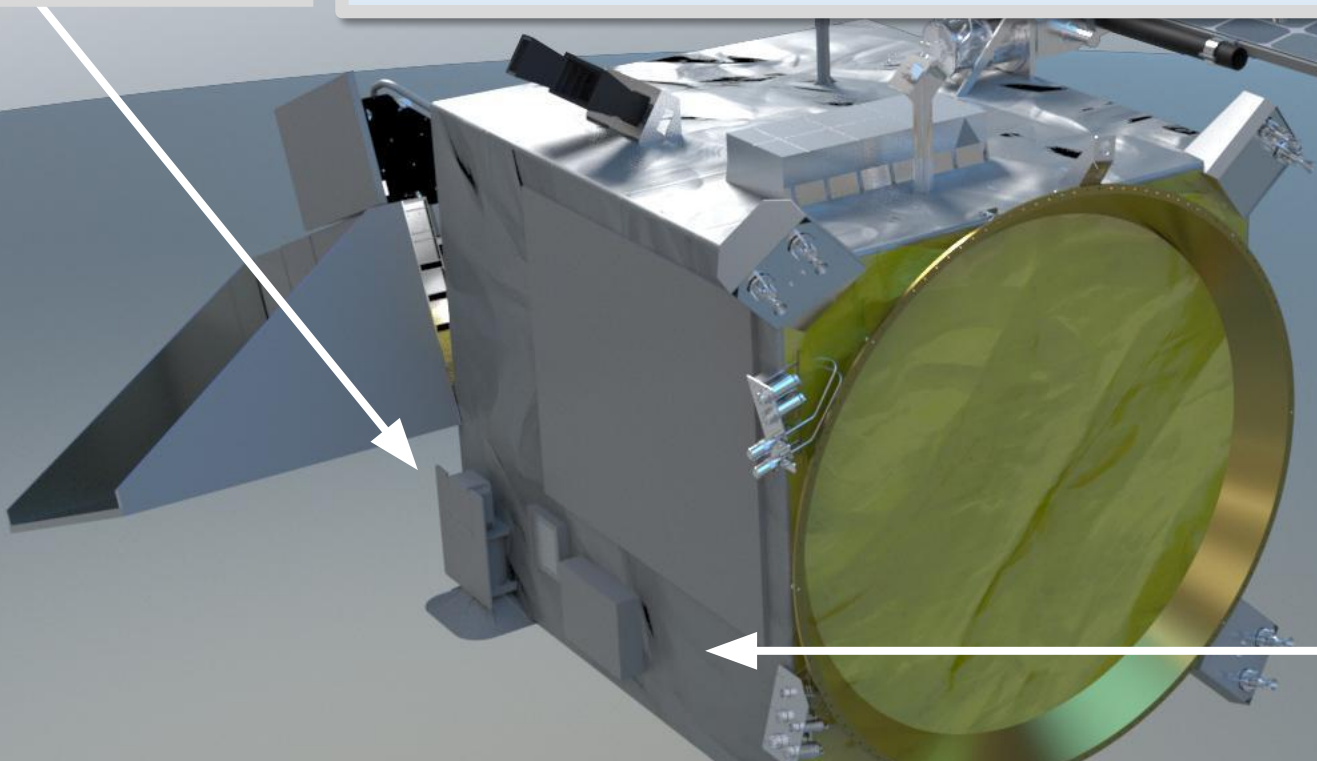


Comparison of the spectral sampling of different missions. Figure from “Dierssen, H. M., Gierach, M., Guild, L. S., Mannino, A., Salisbury, J., Schollaert Uz, S., et al. (2023). Synergies between NASA's hyperspectral aquatic missions PACE, GLIMR, and SBG: Opportunities for new science and applications. *Journal of Geophysical Research: Biogeosciences*, 128, e2023JG007574. <https://doi.org/10.1029/2023JG007574>”

UMBC Hyper
Angular
Rainbow
Polarimeter
(HARP-2)



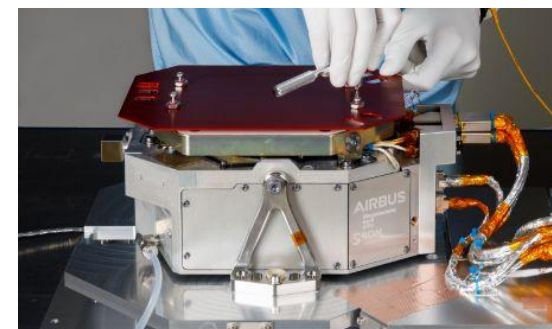
- Excellent for cloud droplet size and ice particle shape/roughness retrievals
- Wide swath matches OCI, offering potentially improved atmos. Correction*
- Physical properties of aerosols: size, shape, distribution vertically and horizontally, how much light do they block (opaqueness)*



Slide modified from, J Werdell

- Aerosol characterization
- What are they made of? Sulfates, dust, smoke, pollution etc*
- How do they affect light? Refraction, absorption, scattering*

SRON/Airbus
Spectro-polarimeter for
Planetary
Exploration
(SPEXone)



Data Products Table

Calibrated Radiometry and Polarimetry | Aquatic Properties to be Produced by OCI | Atmospheric Properties to be Produced by OCI | Land Data Products to be Produced by OCI | Aerosol and Ocean Properties from HARP2 | Aerosol and Land Surface Properties from HARP2 | Cloud Properties from HARP2 | Aerosol, Ocean, and Land Properties from SPEXone | Aerosol and Ocean Properties from OCI + HARP2 + SPEXone

Points of contact (POCs) under Additional Info refer to the PACE Project Science Lead responsible for shepherding an approach through implementation.

Access to data varies with its status (data maturity level). Provisional data are available through [Earthdata Search](#), the OB.DAAC [File Search](#) and [Level 3 & 4 Browser](#). Test and Diagnostic data are available through the OB.DAAC [File Search](#) and [Level 3 & 4 Browser](#). See also “[Access PACE Data](#)”.

What do colors in the “Availability” column mean?

Available

Coming soon!

Currently implementing
and evaluating

No approach currently
identified

Calibrated Radiometry and Polarimetry

Calibrated and geolocated radiometry and polarimetry as observed at sensor.

Product	L2 Suite	Description and Use	Units	Availability	Status	Additional Info
Spectral top-of-atmosphere radiances from OCI	N/A	Spectral radiance observed at the top of the atmosphere.	$\text{W m}^{-2} \text{um}^{-1} \text{sr}^{-1}$	Level-1B 1.2-km native; daily - Level-1C ; 5.2-km; daily	Provisional	Level-1C draft data format and examples
Spectral top-of-atmosphere radiances and polarimetry from SPEXone	N/A	Spectral radiance and polarimetry observed at the top of the atmosphere, for all sensor viewing angles.	Various	Level-1B 5.2-km; daily - Level-1C ; 5.2-km; daily	Provisional	Level-1C draft data format and examples
Spectral top-of-atmosphere radiances and polarimetry from	N/A	Spectral radiance and polarimetry observed at the top of the atmosphere.	Various	Level-1B 5.2-km; daily - Level-1C ; 5.2-km; daily	Provisional	Level-1C draft data format and examples

Levels

Level 1: Calibrated raw data from sensors. Top of atmosphere radiances from OCI, HARP2, and SPEXone. OCI measures Rrs (Remote Sensing Reflectance): color intensity of the ocean when viewed from above.

Level 2: Derived geophysical variables at the original sensor resolution. Products like remote sensing reflectance (Rrs), absorption properties (derived from Rrs), aerosol properties, cloud properties, chlorophyll estimates (derived from absorption), phytoplankton composition

Level 3: Gridded global maps of those variables, averaged over time (daily, monthly, etc.)

Products

Absorption coefficients: how light is absorbed *within* the water by different substances (like phytoplankton or dissolved organic matter). Derived from R_{rs}

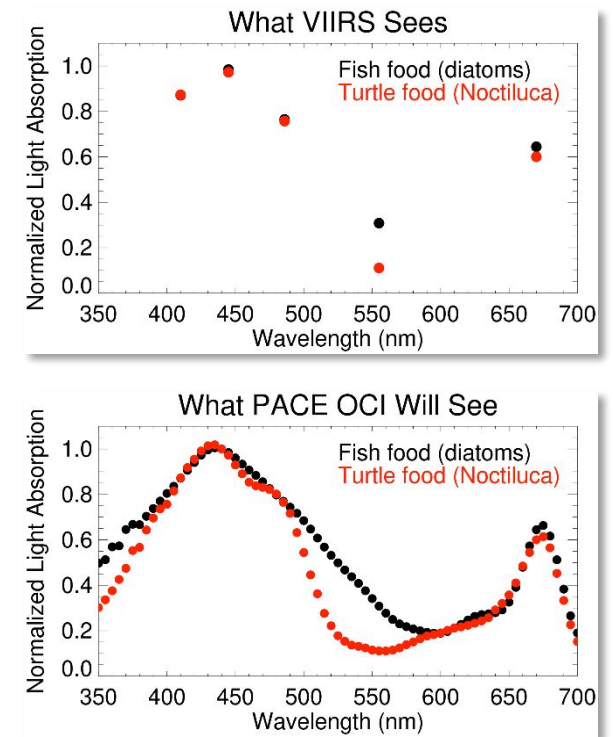
Chlorophyll-a: estimate from empirical equations involving R_{rs} or absorption coefficients

Phytoplankton species and community composition: new to PACE via the hyperspectral absorption coefficients and backscattering coefficients

Diffuse Absorption coefficients (K_d): how quickly light decreases with depth in water at a given wavelength

Spectral phytoplankton absorption coefficients: how much light is absorbed by phytoplankton pigments in water

Phytoplankton carbon: a more direct indicator of phytoplankton biomass



PACE Products for Fisheries, Ryan Vandermeulen (NOAA Fisheries)

Hyperspectral products for fisheries 🔄 🌐



Home

Overview

Introduction
Hyperspectral ocean color
Ocean color satellites

Hyperspectral products

Hyperspectral products
Chlorophyll-a
Phytoplankton carbon
Diffuse attenuation coefficients (Kd)
Spectral phytoplankton absorption coefficients (aph)
Spectral non-algal particle plus dissolved organic matter absorption coefficients (adg)
Spectral particle backscattering coefficients (bbp)
Photosynthetically available radiation (PAR)

Select Language ▼

Powered by Google Translate

Applications of PACE hyperspectral ocean color data for aquaculture and fisheries management

AUTHOR

Ryan Vandermeulen, Veronica Lance, Clete Otoshi,
Ken Riley, Michelle Tomlinson, and Rebecca Trinh;
Python/R code by Elizabeth Holmes

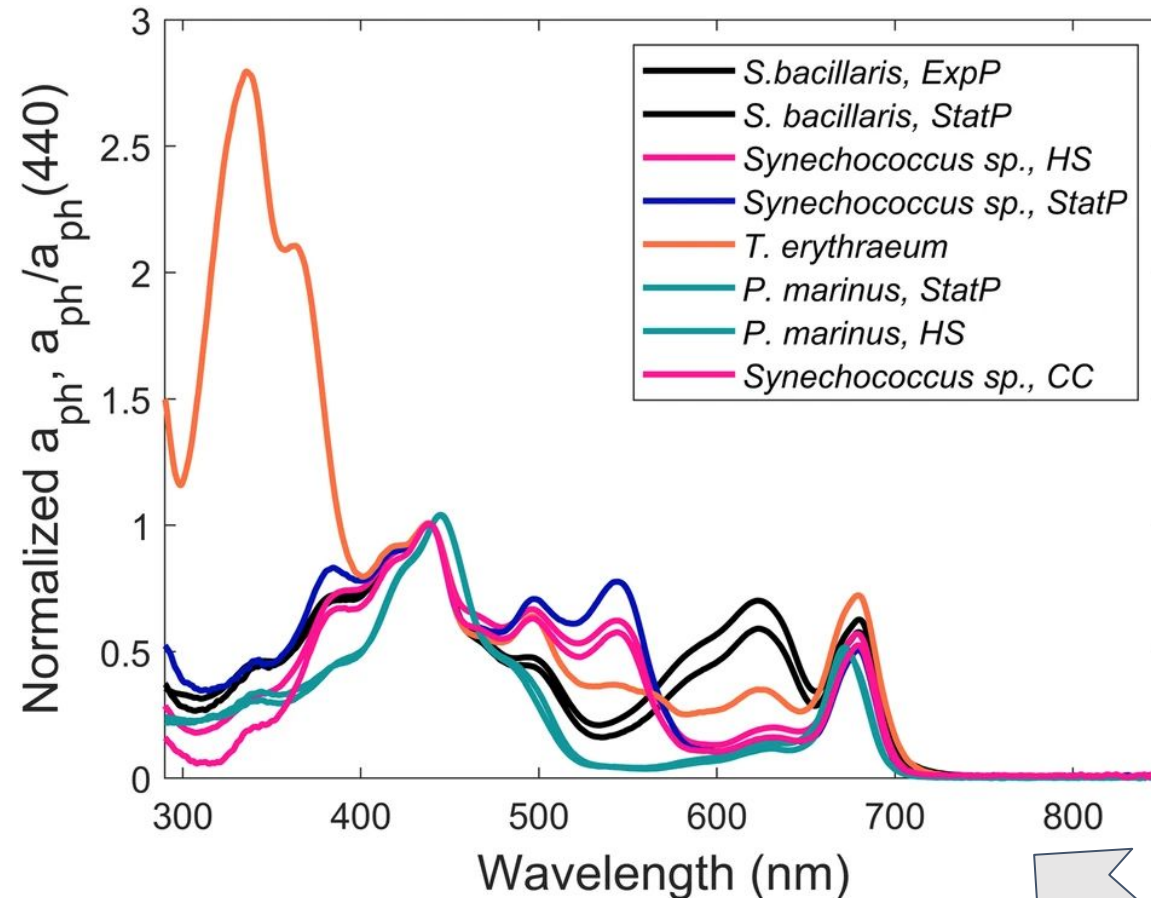
PUBLISHED

2025-04-01T00:00

This is a condensed version of [NOAA Tech. Memo. NMFS-F/SPO-253](#).

PACE Ocean products tell you about:

- WHAT IS THE COLOR OF THE OCEAN (Rrs)
- WHAT/WHO IS THERE (phytoplankton composition, SPM)
- HOW MUCH OF IT IS THERE (chlorophyll concentration, particulate/phytoplankton carbon etc.)
- ARE THEY SHINY/COLORFUL??? (IOPs)
- HOW MUCH LIGHT IS THERE (PAR/Kd)
- WHAT ARE THEY DOING (Primary productivity)
- HOW ARE THEY DOING (Fluorescence Line Height)



Review

Phytoplankton composition from sPACE: Requirements, opportunities, and challenges

Ivona Cetinić ^{a b 1}, Cecile S. Rousseaux ^{b 1} ✉, Ian T. Carroll ^{c b}, Alison P. Chase ^d, Sasha J. Kramer ^e, P. Jeremy Werdell ^b, David A. Siegel ^f, Heidi M. Dierssen ^g, Dylan Catlett ^h, Aimee Neeley ^{b i}, Inia M. Soto Ramos ^{a b}, Jennifer L. Wolny ^j, Natasha Sadoff ^{b i}, Erin Urquhart ^{b i}, Toby K. Westberry ^k, Dariusz Stramski ^l, Nima Pahlevan ^{m i}, Bridget N. Seegers ^{a b}, Emerson Sirk ^{b i}, Priscila Kienteca Lange ^{n o} ...Michael Sayers ^t

[nature](#) > [scientific data](#) > [data descriptors](#) > [article](#)

Data Descriptor | [Open access](#) | Published: 03 February 2024

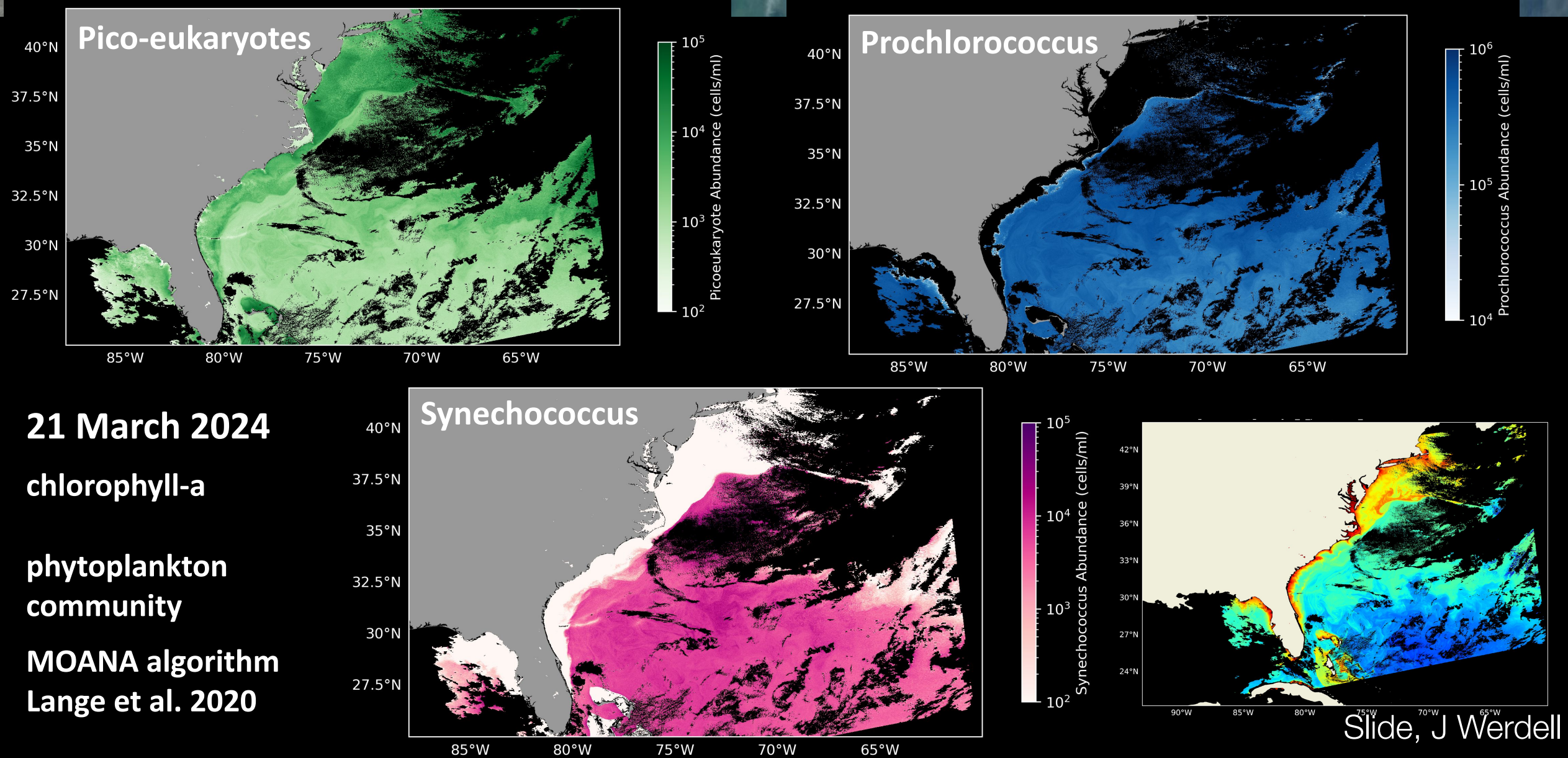
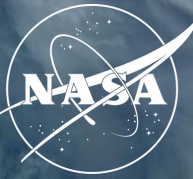
Phytoplankton optical fingerprint libraries for development of phytoplankton ocean color satellite products

Michael W. Lomas ✉, Aimee R. Neeley, Ryan Vandermeulen, Antonio Mannino, Crystal Thomas, Michael G. Novak & Scott A. Freeman

[Scientific Data](#) **11**, Article number: 168 (2024) | [Cite this article](#)

3118 Accesses | 4 Altmetric | [Metrics](#)

PACE OCI Phytoplankton Community Composition



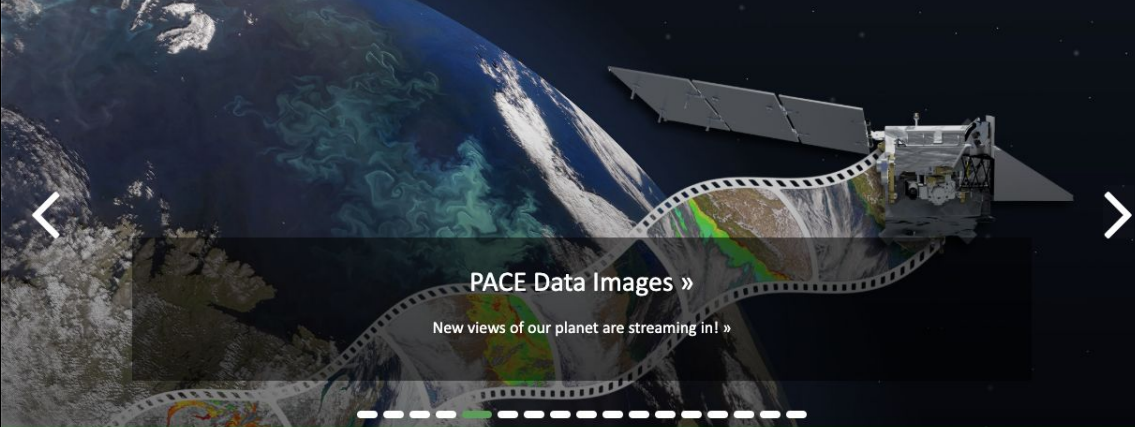
<https://pace.gsfc.nasa.gov>

 National Aeronautics and Space Administration

PACE Plankton, Aerosol, Cloud, ocean Ecosystem



HOME ABOUT MISSION SCIENCE APPLICATIONS DATA LEARN MORE NEWS EVENTS GALLERY DOCUMENTS



PACE Data Images »

New views of our planet are streaming in! »

MISSION ELAPSED TIME	DAYS 423	HR 16	MIN 47	ORBIT NUMBER 944	APPROXIMATE POSITION	LAT 24	LON 109	SATELLITE LIGHTING	DAY	CLICK TO SEE ON MAP
										

✿ PACE HAS LAUNCHED!

Post-launch Updates

Satellite Status Updates

Join the PACE Community Mailing List

PACE's data is helping us better understand how the ocean and atmosphere exchange carbon dioxide. In addition, it is revealing how aerosols might fuel phytoplankton growth in the surface ocean. Novel uses of PACE data will benefit our economy and society. For example, it will help identify the extent and duration of harmful algal blooms. PACE is extending and expanding NASA's long-term

WHAT'S NEW

02-Apr-25
New PACE Terrestrial Data Users' Group

08-Feb-25
PACE-iversary #1

08-Feb-25
A Year of PACE Discoveries

06-Feb-25
PACE-PAX Overview Page

<https://oceancolor.gsfc.nasa.gov/resources/docs/tutorials/>

 OCEAN COLOR
OB.DAAC | OBPG



ABOUT DATA RESOURCES TOOLS COMMUNITY GALLERY FORUM

Quick Links



RESOURCES

HELP HUB

Satellite data processing can be difficult.
We're here to help you climb out of that hole!

We are constantly working on new material. Come back often!

Have an idea? Share it with us on the Earthdata Forum using the tags OB.DAAC under DAAC and Data Recipes under Services/Usage.

Explore our learning resource themes:

- Help Hub Core
- ARSET Training - Introduction to Plankton, Aerosol, Cloud, Ocean Ecosystem (PACE) Hyperspectral Observations for Water Quality Monitoring
- PACE Hackweek 2024
- SeaDAS Basics

Interactive tutorials

Run the code as you follow the training with Jupyter notebooks.

Learn more about Jupyter Notebooks

Cloud computing

Some tutorials use cloud computing resources.

Learn more about the cloud

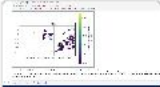
Help Hub Core

PACE MODES Python Interactive

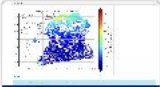
Learn the essential things you need to know to access and process data in this series of Jupyter Notebooks.



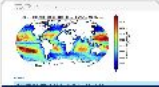
Cloud Data Access (OCI Example)



File Structure (OCI Example)



Explore Level 2 Ocean Color Data



Explore Level 3 Ocean Color Data



Installing the SeaDAS Command Line Tools (OCSSW)



Processing with the SeaDAS Command Line Tools (OCSSW)

Accessing PACE Data

Earthdata Search OB.DAAC portal

Earthdata Search OB.DAAC portal

<https://search.earthdata.nasa.gov/search>

Top left click “Browse Portals”

Click on “OBDAAC”

Filter Instruments to “OCI”

OB.DAAC Search page

<https://oceandata.sci.gsfc.nasa.gov/>

How can I access PACE data that's in the cloud?

You can directly download netcdfs and run analysis. But these files are really big...often better to just get the subset of data that you want.

- OPeNDAP servers allow this so you download or access what you need
- or we can work with data in cloud buckets using earthaccess Python package

Resources

- Applications of PACE hyperspectral ocean color data for aquaculture and fisheries management, R Vandermeulen et al.
- PACE HackWeek 2024 <https://pacehackweek.github.io/pace-2024/>
- ARSET Course
https://appliedsciences.nasa.gov/sites/default/files/2024-10/PACE_Part2_Final_0.pdf
- NOAA HackDays <https://nmfs-opensci.github.io/NMFSHackDays-2025/>
- EDMW PACE Workshop R Vandermeulen and E Holmes